

Spatialphonic360: Acoustic Space for Arbitrary Sound Image Presentation Based on Both Ears Tracking

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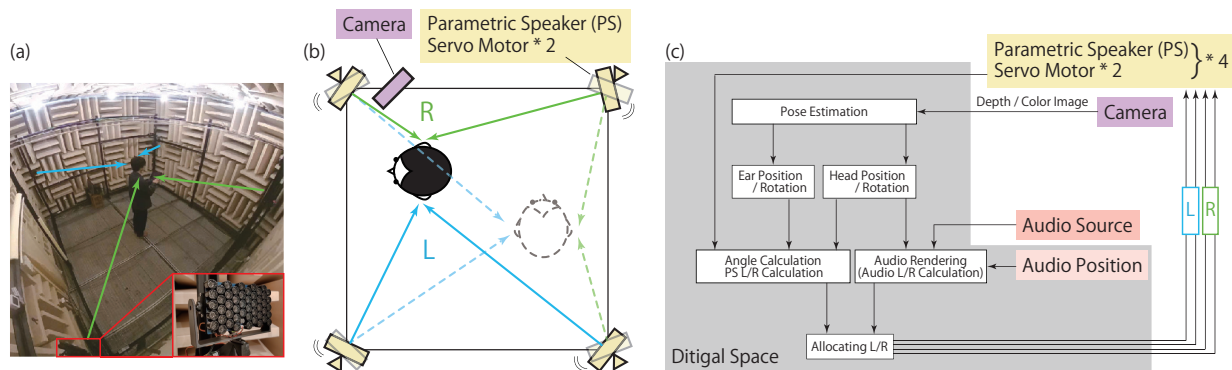


Figure 1: (a) In this study, we construct a system that can present stereophonic sound at any position in the room and any head direction, (b,c) Two motors attached to the parametric speaker rotates according to the listener's position, which is calculated by the camera. A diagram of these system configurations is shown in (c).

ABSTRACT

Conventional stereophonic sound fixed the sweet spot or limited only the front sound image. Therefore, there is a problem that it cannot cover interactivity, such as the listener's movement or turning around. In this study, we developed an interactive stereo system that can present a 360° sound image in any position and direction regardless of the direction of the listener's face. User experiments were conducted in an anechoic chamber at representative points and head directions. The results show that the system can present the sound image with a maximum perceived error of 32.6° and 23.1° for the average, maintaining approximate accuracy in presenting the sound image.

CCS CONCEPTS

• **Human-centered computing** → **Sound-based input / output;**
User studies; Mixed / augmented reality.

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KEYWORDS

stereophonic sound, parametric speaker, human pose estimation

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1 INTRODUCTION

Visual digital content such as augmented reality and virtual reality are becoming increasingly popular. In line with this trend, auditory content is also required to have rich expressions such as stereophonic sound. Wireless headphones may be used as a method of playing stereophonic sound. However, given the difficulty of connection and the ease of interacting with others in a public space, speakers are desirable. As an existing technology, a surround system (e.g., 5.1ch, 7.1ch, etc.) can present sound images in all directions, enabling richer expression than conventional stereo sound sources [2]. However, since this system uses fixed loudspeakers, the sweet spot is limited, making it impossible to listen at any position. In addition,

existing surround sound systems do not support the use of different facial orientations, making a lack of interactivity.

On the other hand, prior researches have developed the stereophonic sound presenting at any position [1, 3]. These are based on using a pair of parametric speakers, which produce narrative directional sound, and stereo sound images can be presented to specific individuals. However, it has only been evaluated in a limited forward direction. Therefore, the presentation of sound images to the rear has not been realized. Besides, it does not cover the changes in facial orientation and remains lack of interactivity.

In this study, we construct an interactive stereo system that can present a sound image at any position and direction regardless of the face orientation. We will also evaluate the accuracy of the sound image localization through experiments. This system will contribute to expanding the flexibility of expression as a stereophonic sound presentation system that is not limited by location.

2 PROPOSED SYSTEM OVERVIEW

As shown in Fig. 1, parametric speakers (Switch Science: SSCI-018425) are installed at the room's four corners so that 3D sound can be heard from anywhere in the room. Half of the four parametric speakers present sound to the left ear and the others to the right ear. A pair of servo motors (FEETECH, FT1117M) are attached to the base of each parametric speaker to move in the azimuth and elevation directions, respectively. A camera installed in one corner is used to estimate the position of the human head and ears. Although the angle of view of the cameras is limited (Field of Illumination: $75^\circ \times 65^\circ$), it is possible to cover the entire room by installing cameras in two corners of the room to compensate for blind spots.

The combination of the parametric speakers and the left and right ears is determined based on the estimated ear positions to avoid the sound presented by parametric speakers being blocked by the head. The displacement angle of the motor is calculated and controlled so that the parametric speakers face the ears. The sound source is converted to stereophonic sound by convolving the head-related transfer function (HRTF) based on the relationship between the direction of the head position and the sound image to be presented. These controls are always performed in real-time, and the system was implemented using Touch Designer. These system flows are shown in Fig. 1 (c).

3 USER EXPERIMENT

The experiment was conducted in an anechoic room to eliminate disturbance and reverberation effects as much as possible. A schematic diagram of the experiment is shown in Fig. 1 (a,b). The participant's listening positions are the three points shown in Fig. 2. By evaluating these positions, it is thought that's the whole image can be evaluated regarding the camera and the room's symmetry. At each position, the participants are asked to face four directions (front, back, left, right) and evaluate the sound image position at each of them.

As an evaluation, participants were asked to listen to the moving sound image for 30 seconds while continuously swiping the perceived sound image location on the iPad. As shown in Fig. 2, the

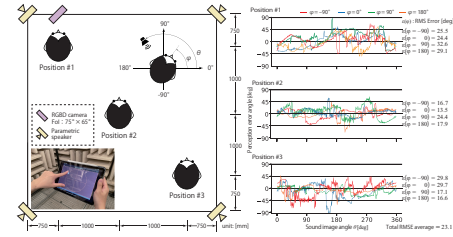


Figure 2: Experimental environment and results: Participants listened to stereophonic sounds at three locations, Position #1–3, facing four directions each. They swiped their iPads to report the direction of the sound they heard. The graph shows the error between the participant report and the actual angle plotted by position and direction. The average RMSE was 23.1° and was almost entirely correct for the direction.

participants are asked to draw the perceived direction with their fingers in a participant-centered interface. The depiction is processed as polar coordinates, and only the angles are extracted. The degree to which this reported angle differs from the actual presented angle is analyzed. The distance from the center shall not be considered. Besides, since there is an unavoidable delay between the sound presentation and participants' drawing, in analysis applies an offset for the time difference.

Since pink noise has been used to test the overall sound localization [1], it is also used as the sound source in this experiment. The direction of the presented sound image moves randomly around the participant to cover all 360° . However, if the sound image position moves too fast, the participants might be unable to keep up with inputting the perceived sound image position. Therefore, we designed a path design algorithm with constraints on the speed of sound image movement and path randomness. The number of participants was five (male: two, female: three), all university students.

The error and root mean squared error (RMSE) of the angular evaluation at each position and face direction are shown in Fig. 2. The maximum of RMSE was 32.6° , and the average was 23.1° for all positions and directions. Thus, it can be shown that the desired sound image localization is approximately possible. Note that there are some points where $RMSE \approx 30^\circ$ for $\varphi = 90^\circ$, $\varphi = 180^\circ$ in Position#1 and $\varphi = -90^\circ$, 0° in Position#3. It seems that the uniformity of the volume from each speaker at the listening point is not sufficient due to the radiation characteristics of the speaker. A correction process for uniformity may be necessary. Although the camera in this system only estimates a person's posture, one concern is the issue of privacy. It is needed to take countermeasures such as using security cameras or other publicly equipment. In addition, it is necessary to test the usability of this system, including the immersive level when it is applied to existing visual digital content. In conclusion, the proposed system provides an interactive stereophonic sound system that can present sound images in any position and any direction (15 directions), regardless of the direction of the face.

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